Gestational Glucocorticoids do not impact fence lizard offspring cell migration

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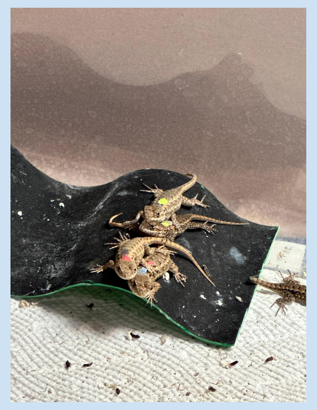
INTRODUCTION

- When exposed to environmental stressors or challenges, the body releases glucocorticoids as a coping mechanism[7]
- Long term exposure to these stressors can cause physiological disruptions such as:
  - cell deterioration[4], disrupted growth patterns[4], decreased wound healing[8]
- Fibroblasts respond to hormones, chemicals, and other stimuli and are integral in wound healing by migrating to and closing the wound[13]
- Previous studies have shown that elevated glucocorticoid levels during gestation can also impact offspring physiology[7]
- By altering the glucocorticoid levels of mothers within ecologically relevant levels[17], we can look at the cellular impacts of glucocorticoids on offspring fibroblast movement.

Figure 1: Female Western Fence Lizard dorsal (left) and hatchlings (right).



HYPOTHESIS

- Maternal stress will increase overall activity in CORT treated offspring fibroblasts.
  - Maternal stress will accelerate fibroblast migration.

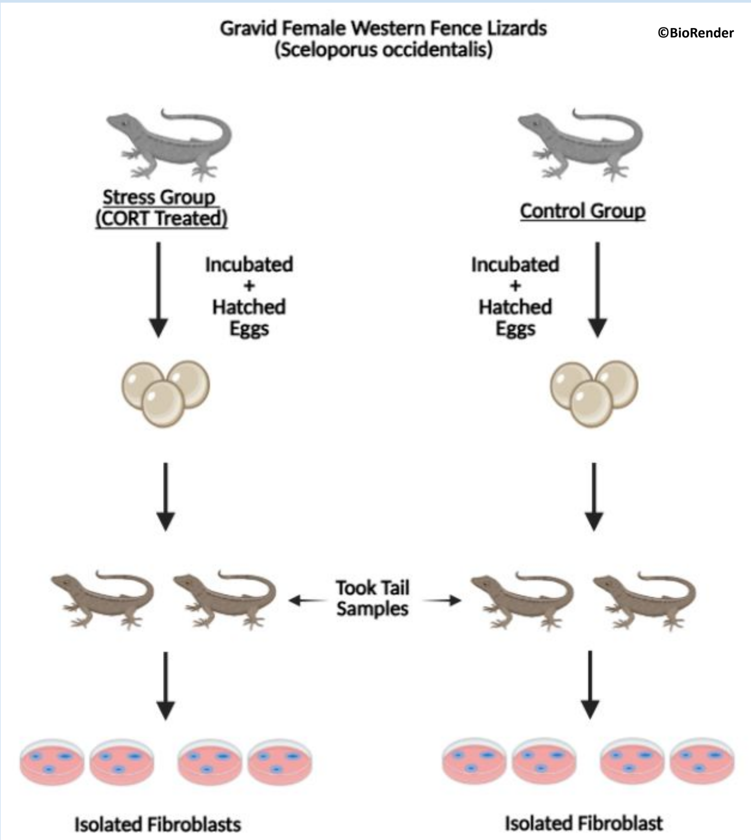


Figure 2: Preliminary methods showing how we obtained offspring fibroblasts.

RESULTS

- No difference was found in gap width closure time between the maternal glucocorticoid and control condition ($F_{1,2.99}$= 0.005, p=0.947; Fig.3)
- Both groups showed a decrease in width as time increased ($F_{1,134}$= 121.979, p<0.001, Fig. 3)

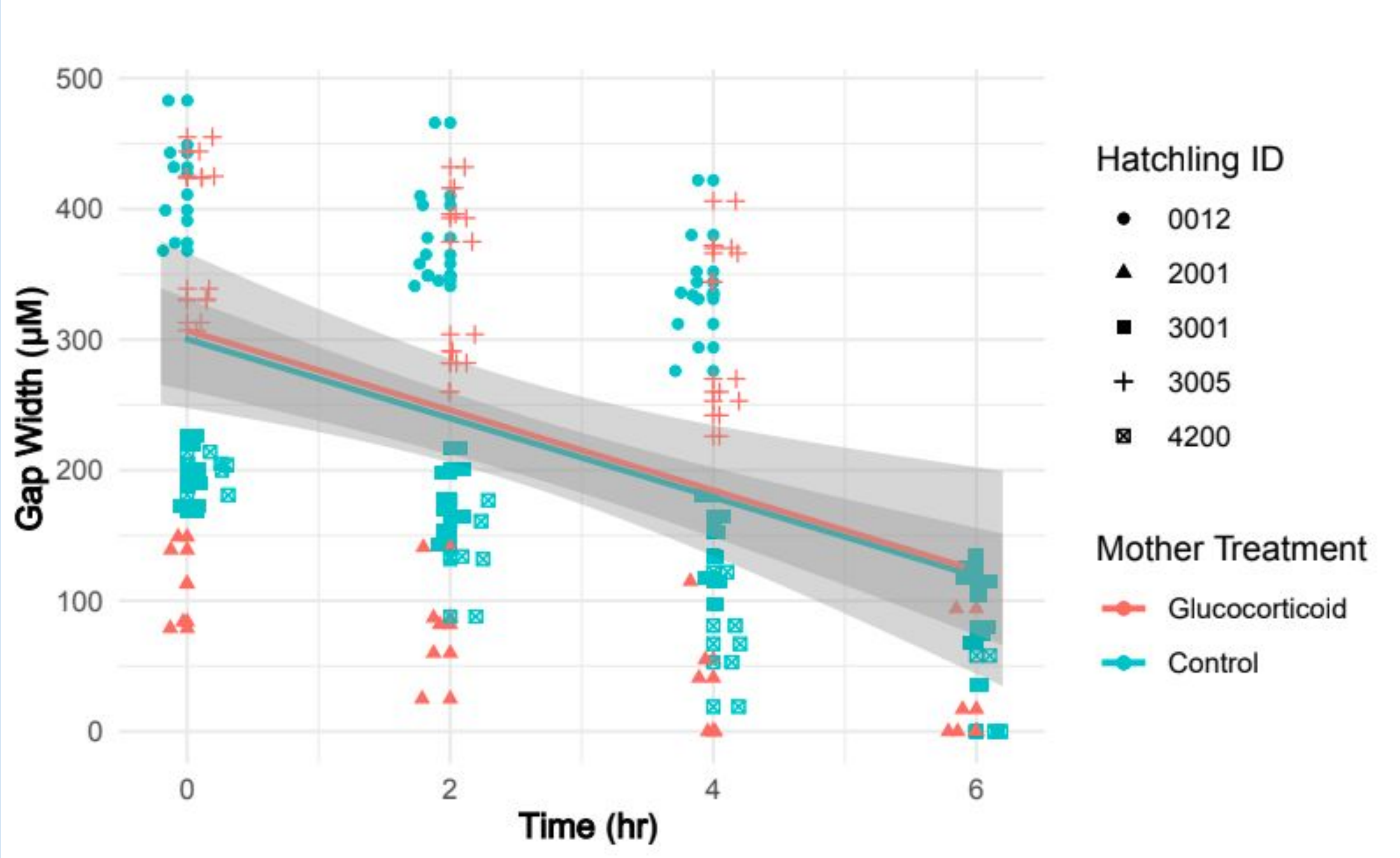


Figure 3: Scatterplot of wound gap width by time in hours showing a decrease in width as time increased.

DISCUSSION

- Offspring from CORT- treated mothers do not show evidence of increased wound healing capacity via increased fibroblast migration rates
  - However, these results are preliminary due to small sample size
- Wound healing may exhibit less plasticity than other traits (ex: hormones, heart rate, and morphology)[7]
- Further research is needed to elucidate the physiological basis of the differences

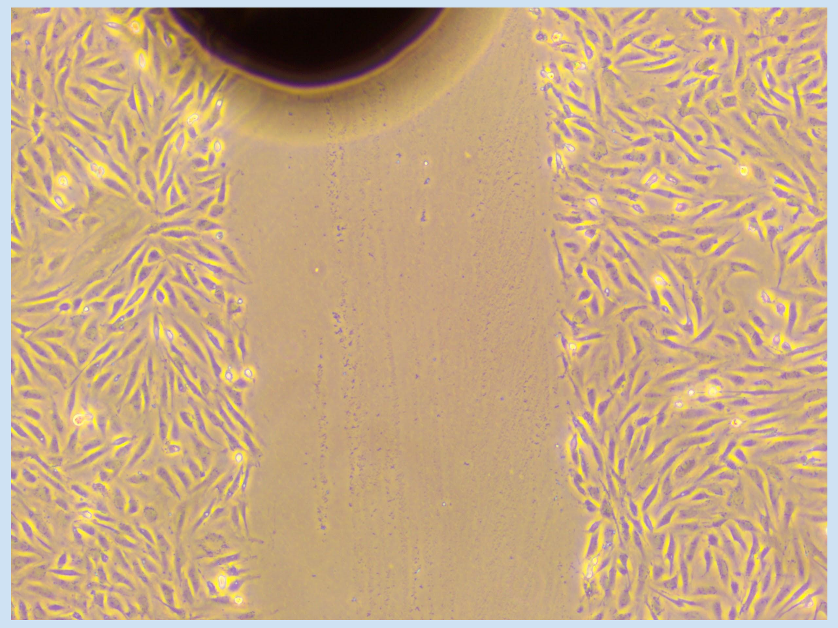


Figure 4: Wound created in cells using a 200µl pipette tip under 40x magnification

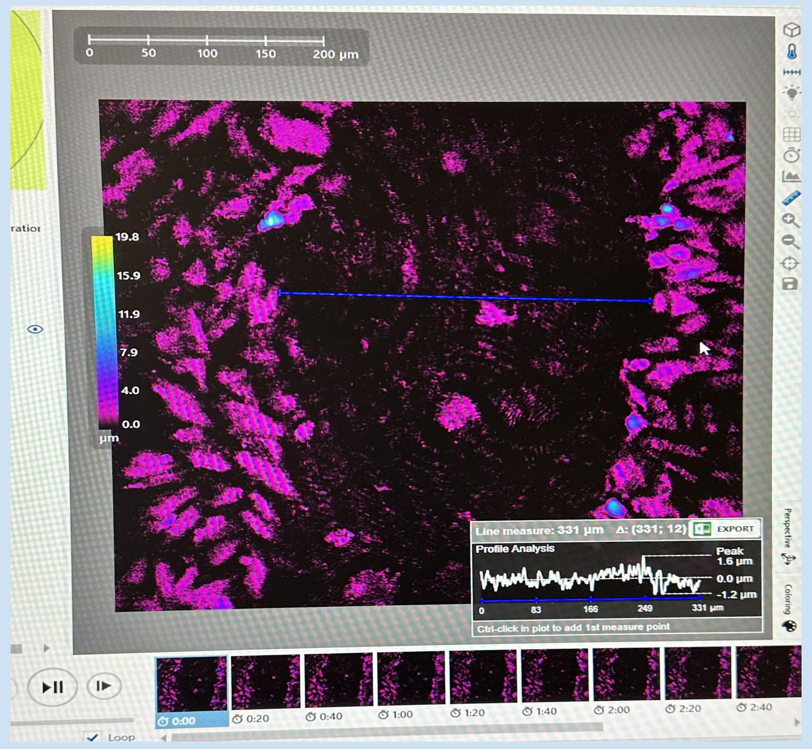


Figure 5: HoloMonitor software used to measure wound distance in µm under 40x magnification

METHODS

Cell Culture:

- Cultured offspring fibroblast of glucocorticoid treated (n=2) and control (n=3) mothers into gelatin-coated dishes
- Maintained cells in complete lizard media

Scratch Assay:

- Passaged fibroblasts and seeded into 6-well plate to reach 100% confluence after 48 hours.
- Made scratch using 2 uL pipette tip through center of wells
- Immediately placed plate into incubated holographic microscope.
- Captured images ever 20 minutes over 24 hour period.
- Analyzed images with Holomonitor Live Cell Imaging System
- LMM for statistical analysis

Future Directions:

- Investigate the impact of maternal stress on offspring lipid peroxidation in fibroblasts[1]
- Assessing alterations to white blood cell proportions from maternal stress[7]

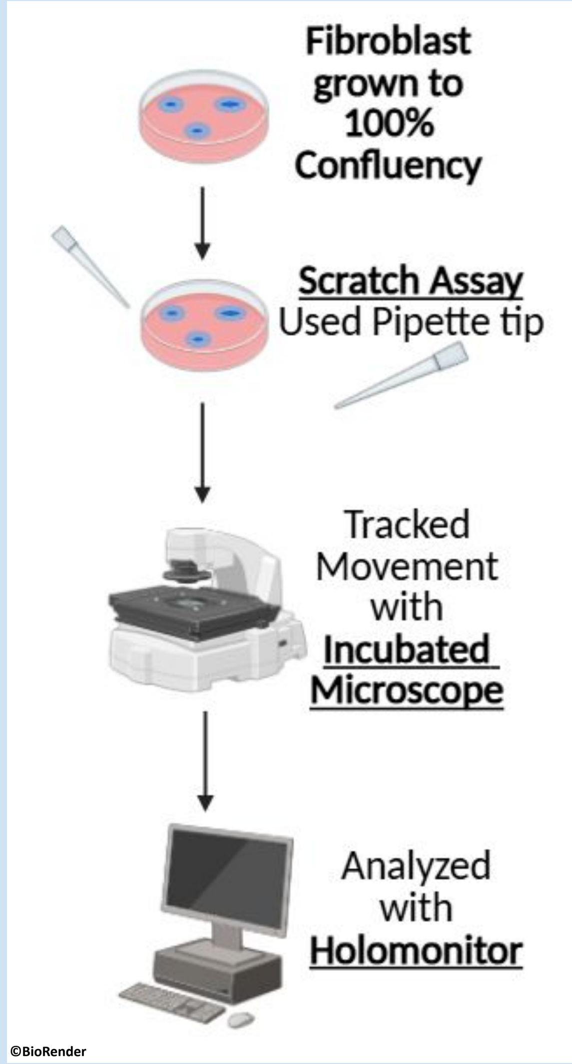


Figure 6: Diagram of methods from fibroblast growth to image analysis

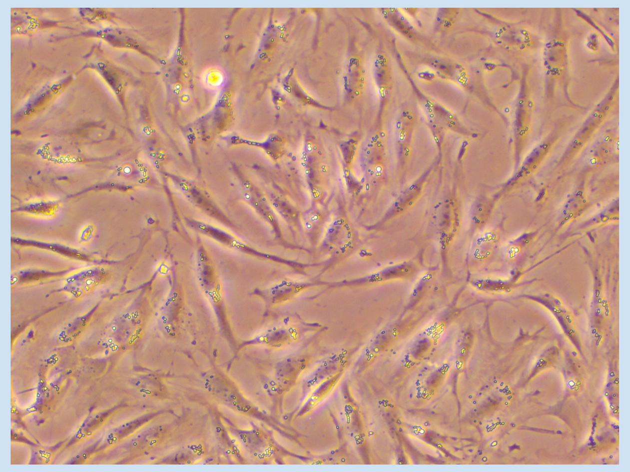


Figure 7: Lizard fibroblasts immediately after treating with RSL-3

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